TAMIL MANI B

Software Engineering · Cloud Computing · Applied AI

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EDUCATION

R.M.K. College of Engineering and Technology, Chennai — Expected May 2028

B.E. Computer Science and Engineering | CGPA: 7.68 / 10

Coursework: Data Structures & Algorithms (C++), OOP (Java), DBMS, Operating Systems, Distributed Systems, Web Development (MERN).

School: HSC 86.16% (Velammal Matriculation HSS, 2024). SSLC 84.80% (Leo Matriculation HSS, 2022).

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript (ES6+), TypeScript, SQL, Embedded C/C++

Cloud & DevOps: Microsoft Azure (AZ-900), Azure Resource Manager APIs, Entra ID / RBAC, Docker, CI/CD, REST APIs

Web & Backend: Next.js 14, React 18, Node.js, Express, Flask, FastAPI, SQLAlchemy, Tailwind CSS, Supabase

AI & Data: TensorFlow.js, TensorFlow Lite, scikit-learn, LightGBM, SHAP, NetworkX, Groq (Llama-3.1)

Systems & Tools: WebSockets, MQTT, ESP32, Arduino, IndexedDB, Twilio Voice, Git, GitHub, Postman, Linux CLI

SELECTED ENGINEERING PROJECTS

CloudPulse | Cloud Cost Monitoring and Optimisation Platform (team of 4) — Azure · FastAPI · Python

- Built a read-only Azure platform (service principal with Reader and Cost Management Reader roles) that detects idle disks, unused public IPs, orphaned snapshots and idle or oversized VMs, and scores each finding P1–P4.
- Implemented a weekday-aware robust z-score for spend anomalies, Holt smoothing for budget forecasts and 7-day billed-cost verification of savings; validated with 27 automated tests.

SENTINEL-G | Explainable AML and Mule-Account Platform — **Top 10 Finalist, IOB FinTech Hackathon**

- Architected a fraud-detection platform combining 7 deterministic AML engines (velocity, structuring, PEP, KYC) with graph ML, using NetworkX cycle detection, PageRank and betweenness centrality fused with Isolation Forest and LightGBM scoring, explained with SHAP.
- Designed evidence governance with AES-256-GCM, SHA-256 hash-chained audit logs and Ed25519-signed maker-checker workflows.

NalamMesh | Resilient Rural Public Health Platform — **Aithon 2.0: 4th Runner-up (₹4,000) · SIH 2026**

- Designed an offline-first healthcare PWA linking Sub-Centres → PHCs → CHCs → SDHs → District Hospitals with ABHA-linked records and ABDM / FHIR R4 interoperability.
- Built an in-browser TensorFlow.js model for 8-vital edge triage that works on 2G or zero connectivity, with IndexedDB sync.

H2 Eco | Edge-AI Hydrogen Fuel Optimisation System — **College Seed Funding (₹2.75 lakh)**

- Built a bench-tested SS316L electrolyzer prototype with an ESP32 edge-AI controller (TensorFlow Lite) and an MQTT IoT safety layer; recorded 15–25% fuel reduction and 20–40% emission decrease in engine test-bench trials.

Aegis Medical | AI Voice Receptionist and Revenue OS — Node.js · Twilio · Groq · Supabase

- Built a real-time phone receptionist on Twilio Voice with sub-800 ms responses (Groq Llama-3.1, ElevenLabs) and an LLM pipeline that extracts ICD-10/CPT codes and triage priority from call transcripts into Supabase.

LEADERSHIP & TECHNICAL ENGAGEMENT

BUGQUEST'26 | Lead Organizer, Interdepartmental Debugging Contest — IEEE Day, R.M.K.C.E.T. · Oct 2026

- Organised a college-wide contest with 61–100 participants from multiple departments, coordinating faculty and volunteers to run timed coding rounds.
- Designed 12 debugging problems across 3 rounds in C++, Java and Python, each with automated test cases, to build participants' debugging and problem-solving skills.

ACHIEVEMENTS & PUBLICATION

- **Aithon 2.0 Hackathon:** 4th Runner-up with ₹4,000 cash prize (NalamMesh). **IOB FinTech Hackathon:** National Top 10 finalist (SENTINEL-G).
- **Research:** Co-authored "Neural Architecture Search (NAS) Optimization for Deep Learning Models", presented at NCECTISS-25 National Conference (2025). **Microsoft Imagine Cup 2026:** Participant.

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)**, July 2026. Credential ID 2A513E569A92A1A0.
- **JPMorgan Chase & Co. Software Engineering Job Simulation** (Forage), Jan–Feb 2026: Kafka integration, H2 in-memory DB, REST APIs.